

HAIYUN DENG

Miami, FL 33199 | haiyundeng@outlook.com | 786-932-8826 | haiyundeng.com

linkedin.com/in/haiyun-deng | github.com/hd007

Summary

Ph.D. researcher in Computer Science specializing in AI security, privacy, and applied machine learning, with industry experience in backend and API development. Proven experience designing large-scale data analysis pipelines, building ML-based detection models, and conducting security and privacy evaluations across Android, healthcare, blockchain, and satellite communication systems. Published author in top-tier security venues with hands-on experience applying LLMs to real-world security workflows.

Education

Florida International University, Doctor of Philosophy (Ph.D.) in Computer Science Aug 2023 – Dec 2027

- GPA: 4.0/4.0

- **Coursework:** Theory of Computation, Analysis of Algorithms, Advanced Data Mining, Artificial Intelligence

Florida International University, Bachelor of Science (B.S.) in Computer Science May 2021 – Aug 2023

- GPA: 4.0/4.0

- **Coursework:** Computer Architecture, Operating Systems, Database Systems, Net-centric Computing

Technologies & Competencies

Software Engineering: Backend Development • REST APIs • Spring Framework • Microservices • Database Design (SQL/NoSQL) • Docker & Cloud Deployment (AWS/Azure) • CI/CD Pipelines (Jenkins)

Machine Learning & AI: LLMs • PyTorch/TensorFlow • Seq2Seq Models (LSTM/RNN) • Feature Engineering • Classification & Detection Systems • Model Evaluation & Optimization

Security & Privacy: Application Security Analysis • Vulnerability Detection • Privacy Compliance (HIPAA/GDPR/COPPA/CCPA) • Cryptographic Systems • Threat Modeling

Languages & Tools: Python • Java • C/C++ • JavaScript • SQL • Pandas/NumPy • Wireshark • Git • Linux/Unix

Technologies

Languages: Python, Java, C/C++, C#, JavaScript, SQL

Machine Learning & AI: LLMs, PyTorch, TensorFlow, Scikit-Learn, CNNs, RNNs, Pandas, NumPy

Cloud & DevOps: AWS, Azure, Docker, Jenkins

Backend & APIs: REST APIs, Spring Framework

Databases: MySQL, MongoDB

Security & Privacy: Wireshark, Exodus, Android Security, Blockchain Security

Compliance: HIPAA, GDPR, COPPA, CCPA, PDPA

Generative AI: Text-to-Speech (TTS), Speech-to-Text (STT)

Experience

Graduate Research and Teaching Assistant, Florida International University Aug 2023 – Present

- Designed and deployed large-scale security and privacy analysis pipelines (567K+ Android updates), spanning mHealth compliance, phishing detection, blockchain oracle security, and satellite key management.
- Built ML-based phishing and fraud detection models using sequence modeling and feature learning.
- Led labs and mentored students in systems, security, and networking courses.

Software Engineer Intern, American Express

May 2023 – May 2023

- Developed and tested production-grade REST APIs using Spring Framework within enterprise-scale systems.
- Designed modular, reusable API components to improve maintainability and support future feature expansion.

Computer Architecture Course Grader, Florida International University

Jan 2023 – May 2023

- Graded and evaluated assignments and exams with rubric-based assessment, providing targeted feedback.

Programming Tutor, Florida International University

Sep 2021 – Nov 2023

- Tutored students in Java, C, data structures, and discrete mathematics through hands-on coding sessions.

Publications

- H. Deng et al., "Unveiling the Global Landscape of Android Security Updates," in IEEE Transactions on Dependable and Secure Computing, vol. 23, no. 3, pp. 5971–5988, May–June 2026, doi: 10.1109/TDSC.2026.3660642.
- H. Deng, A. Sahin, K. Akkaya and S. Uluagac, "Adaptive Solutions for DeFi: Leveraging T2EMA's Dynamic Oracle Protection," 2025 IEEE International Conference on Blockchain and Cryptocurrency (ICBC), Pisa, Italy, 2025, pp. 1-5, doi: 10.1109/ICBC64466.2025.11114707.

Projects

Android Security Update Analysis

github.com/android_security

- Designed and implemented a large-scale analysis pipeline covering 567K+ Android security updates across 904 devices to quantify update delays, OEM patching behavior, and systemic security exposure.
- Identified longitudinal update patterns and risk disparities across manufacturers, enabling empirical evaluation of Android ecosystem security practices.
- **Tools Used:** Python, Pandas, NumPy, Jupyter Notebook

Privacy Analysis of mHealth Apps

- Conducted static and dynamic privacy analysis of mHealth applications to uncover data collection, transmission, and sharing behaviors inconsistent with HIPAA, GDPR, and COPPA requirements.
- Leveraged LLM-assisted inspection to scale policy interpretation and privacy risk identification across heterogeneous app behaviors and network traces.
- **Tools Used:** LLMs, Static Analysis, Dynamic Analysis, Network Monitoring

Phishing Domain Detection with Seq2Seq Model

- Developed an LSTM-based Seq2Seq model to detect phishing domains by learning sequential and lexical patterns in domain names.
- Demonstrated robustness against obfuscated and previously unseen phishing domains through sequence-level feature learning.
- **Tools Used:** Python, TensorFlow, RNN, LSTM, Machine Learning

Satellite Communication Security

- Systematically surveyed and categorized key management techniques for satellite communication systems, analyzing trade-offs across latency, isolation, scalability, and security.
- Proposed an improved key management approach addressing identified gaps in existing satellite communication security models.

Secure Authentication in Next-Generation Wearables

github.com/wiot_auth

- Systematized 166 research papers (2007–2024) on wearable authentication methods, introducing the first unified taxonomy spanning six interaction models and 36 authentication attributes.
- Performed comparative evaluation of physiological, behavioral, contextual, hybrid, and cryptographic authentication modalities against security, usability, and scalability requirements.
- Analyzed the evolution of wearable authentication from early biometric systems to multimodal, decentralized, and ecosystem-centric designs.